--- Document: cyber_security(3).pdf | Page 1 ---

MAJOR PROJECT[BCI685]

SYNOPSIS

ON

AI DRIVEN HARDWARE-BASED CYBER SECURITY THREAT DETECTION &PREVENTION SYSTEM WITH IoT INTEGRATION

Submitted in partial fulfillment of requirement for the award of the degree

BACHELOR OF ENGINEE RING

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COMPUTER SCIENCE & ENGINEERING [AI&ML]

SUBMITTED BY

NIKHIL N
4KV23CI035
PRAJNA K
4KV23CI042
VAISHNAV K M
4KV23CI054
VARUNKUMAR R 4KV23CI057

Under the guidance of

Mr. KIRAN B B
Lecturer
Dept. of CS&E(AI&ML)
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING[AIML]
K. V. G. COLLEGE OF ENGINEERING, SULLIA, D.K ? 574 327

--- Document: cyber_security(3).pdf | Page 2 ---

K.V.G COLLEGE OF ENGINEERING, SULLIA, D.K ?574 327
(AFFILIATED TO VISVESVARAYA TECHN OLOGICAL UNIVERSITY, BELAGAVI)
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING[AIML]

MINI PROJECT[BAI586]

AI DRIVEN HARDWARE-BASED CYBER SECURITY THREAT
DETECTION &PREVENTION SYSTEM WITH IoT INTEGRATION

Submitted by,

NIKHIL N
4KV23CI035
PRAJNA K
4KV23CI042
VAISHNAV K M
4KV23CI054
VARUNKUMAR R
4KV23CI057

This project/report has been reviewed and approved by the following

Mr. KIRAN B B
Lecturer
Dept. of CS&E(AI&ML)

Date:

Place:

ABSTRACT

In modern network environments, traditional rule-based security mechanisms such as firewalls and signature-based intrusion detection systems are increasingly insufficient against novel and sophisticated cyber-attacks. This project presents an AI-driven cybersecurity defense system deployed as a physical hardware appliance that combines Machine Learning (ML), Deep Learning (DL), and IoT security monitoring to provide real-time detection, classification, and prevention of cyber threats. The system leverages a Raspberry Pi as the main security gateway that continuously captures live network traffic and IoT device data. Network traffic features are extracted and analyzed using trained ML and DL models to identify malicious behavior patterns including brute force, port scanning, and anomaly-based attacks. Upon detection of suspicious activity, the device executes automatic mitigation actions such as blocking malicious IP addresses, terminating suspicious sessions, and

anomaly-based attacks. Upon detection of suspicious activity, the device executes automatic mitigation actions such as blocking malicious IP addresses, terminating suspicious sessions, and
generating controlled warning responses. Additionally, the project integrates ESP32-based IoT nodes to simulate smart devices and monitor device-level anomalies. A unified web dashboard provides real-time visualization of detected threats, attack types, confidence scores, blocked events, and system metrics. All demonstrations are performed in a controlled and ethical lab environment using intentionally vulnerable applications, ensuring legal and safe operation. This work contributes a practical, low-cost, and scalable hardware prototype capable of emulating enterprise-level intrusion detection and prevention functionality while maintaining ethical cybersecurity standards

1.Introduction

With the rapid expansion of digital technologies, modern networks are increasingly vulnerable to cyber threats such as malware, brute force attacks, port scanning, and unauthorized access. Traditional cybersecurity solutions such as firewalls and signature-based intrusion detection systems rely heavily on predefined rules and known attack signatures. While these methods were effective in earlier network environments, they are no longer sufficient to detect sophisticated and previously unseen cyber attacks.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have opened new possibilities for intelligent cybersecurity systems capable of detecting abnormal network behavior. AI-based systems can analyze large volumes of network traffic, learn patterns of normal activity, and identify potential threats in real time. This enables proactive defense

patterns of normal activity, and identify potential threats in real time. This enables proactive defense
mechanisms rather than relying solely on reactive security approaches.

This project proposes an AI-driven cybersecurity defense system implemented as a hardware security appliance using a Raspberry Pi. The device functions as a network security gateway that continuously monitors network traffic and IoT device activity. By integrating Machine Learning and Deep Learning models, the system can detect suspicious activities such as brute force attempts, port scanning, and anomaly-based intrusions.

Additionally, the system incorporates ESP32-based IoT nodes to simulate smart devices and monitor device-level security events. A web-based dashboard provides real-time visualization of detected threats, system status, and attack analytics. The project aims to demonstrate how a low-cost hardware platform can emulate enterprise-level intrusion detection and prevention systems (IDS/IPS)

threats, system status, and attack analytics. The project aims to demonstrate how a low-cost hardware
platform can emulate enterprise-level intrusion detection and prevention systems (IDS/IPS)
while maintaining ethical cybersecurity practices within a controlled laboratory environment.

1.1Scope

The scope of this project focuses on developing a practical AI-based cybersecurity defense prototype capable of monitoring and protecting network environments from potential cyber threats.

The system will monitor live network traffic and IoT device communications in a controlled lab environment. Machine learning models will analyze extracted network features to identify suspicious activities and classify them into different types of cyber attacks. Once detected, the system will automatically apply mitigation measures such as blocking malicious IP addresses or terminating suspicious sessions.

The project includes the development of a web-based dashboard that provides real-time visualization of network threats, attack categories, confidence scores, and system performance metrics. The system will also demonstrate the integration of IoT devices using ESP32 to simulate real-world smart devices and monitor their security behavior.

The prototype aims to provide a low-cost and scalable cybersecurity solution suitable for

and monitor their security behavior.
The prototype aims to provide a low-cost and scalable cybersecurity solution suitable for
educational institutions, small organizations, and research laboratories. Future enhancements may include support for additional attack detection models, integration with cloud-based security platforms, and deployment in larger enterprise networks.The main objective of this project is to design and implement an AI-based cybersecurity defense system capable of detecting and mitigating cyber threats in real time.

1.2 Objective

The main objective of this project is to design and implement an AI-based cybersecurity defense system capable of detecting and mitigating cyber threats in real time. The project aims to develop a hardware-based cybersecurity monitoring system using Raspberry Pi as the central security gateway to observe network activity continuously. It focuses on capturing and analyzing live

hardware-based cybersecurity monitoring system using Raspberry Pi as the central security
gateway to observe network activity continuously. It focuses on capturing and analyzing live
network traffic in order to detect suspicious activities and potential attack patterns. The system also aims to implement Machine Learning and Deep Learning models to identify cyber threats such as brute force attacks, port scanning, and anomaly-based intrusions. Another objective is to integrate IoT devices using ESP32 modules for monitoring device-level network behavior and identifying anomalies in smart devices. The project further intends to implement automated response mechanisms that can block malicious IP addresses and terminate suspicious sessions when threats are detected. In addition, a web-based dashboard will be developed to visualize detected threats, attack

classifications, system metrics, and security alerts in real time. Finally, the project aims to create a low-cost and scalable prototype that demonstrates enterprise-level intrusion detection and prevention capabilities.

1.3 Methodology

The proposed AI-driven cybersecurity threat detection system is implemented using a Raspberry Pi as the central security gateway to monitor and analyse network traffic. Initially, the Raspberry Pi is connected to the network through a router or switch to capture incoming and outgoing packets. Network traffic data is collected using packet capturing tools such as Scapy. The captured data is then pre-processed to remove noise and extract relevant features such as IP address, port number, protocol type, and packet size.

Machine Learning and Deep Learning models are trained using cybersecurity datasets to identify

type, and packet size.

Machine Learning and Deep Learning models are trained using cybersecurity datasets to identify patterns related to cyber threats such as brute force attacks, port scanning, and anomalous network behaviour. The trained model is deployed on the Raspberry Pi to perform real-time threat detection. ESP32 modules are integrated to monitor IoT device behavior and send data to the central system. When a potential threat is detected, the system automatically performs response actions such as blocking malicious IP addresses and terminating suspicious sessions. A web-based dashboard is developed to display detected threats, attack classifications, system logs, and security alerts. Finally, the system is tested with simulated attack scenarios to evaluate detection accuracy and system performance.

2.Literature Survey

[1] Proposes an AI-based cybersecurity framework for Active Directory environments using machine learning and graph-based techniques. The system analyzes network relationships and attack paths to detect potential threats and classify network security status effectively.

[2] Provides a detailed study of different types of attacks targeting Active Directory systems, including attack stages, signatures, and vulnerabilities. The paper highlights the importance of monitoring authentication patterns and security logs to prevent enterprise network breaches.

[3] Introduces an AI-assisted framework that uses machine learning algorithms to evaluate the security of Active Directory infrastructures. The framework analyzes network graphs and vulnerability relationships to identify potential security threats.

[4] Presents a comprehensive review of machine learning techniques used in cybersecurity

relationships to identify potential security threats.
[4] Presents a comprehensive review of machine learning techniques used in cybersecurity
systems. The study discusses classification and anomaly detection methods that can identify malicious network activities and improve threat detection accuracy.

[5] Focuses on detecting Kerberoasting attacks in Active Directory environments using Support Vector Machine (SVM) models. The method analyzes event logs and authentication patterns to differentiate between legitimate and malicious access attempts.

[6] Proposes a machine learning model for identifying malicious network payloads in software-defined networks. The study uses algorithms such as Decision Trees and Logistic Regression to analyze both encrypted and unencrypted network traffic.

[7] Discusses advanced digital forensic techniques used to detect and respond to security breaches in Active Directory systems. The study emphasizes the importance of real-time monitoring and

[7] Discusses advanced digital forensic techniques used to detect and respond to security breaches in Active Directory systems. The study emphasizes the importance of real-time monitoring and
automated response mechanisms.

[8] Analyses how ransomware attacks affect Windows Active Directory services and enterprise authentication systems. The study highlights the risks associated with privilege escalation and unauthorized network access.

[9] Explores the integration of neural networks, fuzzy logic, and IoT technologies for enhancing cybersecurity systems. The research demonstrates how intelligent systems can detect cyber threats and improve automated security management.

[10] Presents a machine learning-based classification approach using pattern recognition techniques. Although focused on medical data classification, the study demonstrates the effectiveness of ML algorithms for accurate pattern detection in complex datasets.

3.Problem Statement

Traditional network security mechanisms such as firewalls and signature-based intrusion detection systems are limited in their ability to detect new and evolving attack patterns. These systems depend on predefined signatures and cannot easily identify unknown or zero-day threats. Additionally, many small organizations and educational institutions lack access to affordable enterprise-level cybersecurity solutions.

Modern organizations rely heavily on networked systems, cloud services, and IoT devices. These environments generate large amounts of network traffic and introduce new security vulnerabilities. As networks grow more complex, cyber attackers exploit weaknesses through automated tools, malware, and advanced persistent threats.

This project proposes an AI-driven cybersecurity Défense system deployed as a hardware appliance

malware, and advanced persistent threats.

This project proposes an AI-driven cybersecurity Défense system deployed as a hardware appliance using Raspberry Pi. The system uses Machine Learning and Deep Learning algorithms to analyse network traffic in real time and detect suspicious behaviour patterns. Once a potential threat is detected, the system automatically performs mitigation actions such as blocking malicious IP addresses, terminating suspicious sessions, and generating alerts through a web dashboard. This approach provides a smart, automated, and cost-effective network security solution.

6. Conclusion

The AI-Driven Cybersecurity Threat Detection and Prevention System aims to provide an intelligent and real-time solution for detecting and mitigating network attacks. By integrating machine learning techniques with cybersecurity mechanisms, the system continuously monitors network traffic, analyzes packet behavior, and identifies potential threats automatically. The use of Raspberry Pi as a dedicated security gateway enables real-time packet capture, attack detection, and automatic blocking of malicious sources using firewall rules

In addition, the system incorporates IoT monitoring using ESP32, allowing the detection of suspicious activities in IoT environments, which are often vulnerable to cyberattacks. A web-based dashboard and a custom IDS monitoring software provide administrators with real-time alerts,

suspicious activities in IoT environments, which are often vulnerable to cyberattacks. A web-based dashboard and a custom IDS monitoring software provide administrators with real-time alerts,
attack logs, and detailed reports, enabling effective security management and analysis.

Overall, this project demonstrates how Artificial Intelligence, Cybersecurity, and IoT technologies can be combined to build a smart, automated intrusion detection and prevention system. The proposed system enhances network security by reducing response time, minimizing human intervention, and providing a scalable framework for protecting modern digital infrastructure

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